Improvements v1

Critical Issues & Improvements Needed

1. **Scoring Methodology - One Size Doesn't Fit All**

Problem: The current system uses universal scoring ranges across all industries, but financial metrics vary dramatically by sector. A P/E of 15 might be high for a utility but low for a tech company. A debt-to-equity of 0.5 is different for a bank versus a software company.

Recommendation:

- Implement **sector-specific benchmarks** rather than universal thresholds
- Create industry adjustment factors (e.g., Tech, Financial Services, Utilities, Healthcare each get different scoring rubrics)
- Consider adding a "sector percentile" score - how does this company rank within its industry peers?

2. **Missing Critical Metrics**

Your system overlooks several important fundamental indicators:

Should Add:

- **Free Cash Flow (FCF) and FCF Yield** - more reliable than earnings for valuation
- **Cash Conversion Cycle** - measures operational efficiency
- **Gross Margin trends** - not just absolute profit margin
- **Asset Turnover Ratio** - efficiency metric
- **Earnings Quality Score** - are earnings from operations or one-time items?
- **Working Capital Ratio** - short-term financial health
- **ROIC (Return on Invested Capital)** - better than ROE for capital efficiency

3. **Weighting Structure Issues**

Current Problem: The 30-30-20-15-5 weighting is too rigid.

Recommendations:

- For **growth stocks**: Increase Growth weight (30%), reduce Valuation weight (20%)
- For **value stocks**: Increase Valuation (35%), reduce Growth (15%)
- For **income stocks**: Increase Dividend weight (15-20%)
- Consider a **dynamic weighting system** based on company characteristics or user investment style

4. **Handling Missing Data - Current Approach is Flawed**

Problem: Assigning a neutral score of 5 to missing categories is problematic. If a company doesn't pay dividends (common for growth companies), giving it a 5 is misleading.

Better Approach:

- Redistribute the weight of missing categories proportionally to other categories
- For dividends specifically: If no dividend, give 0 score but reduce dividend weight to 1% and redistribute the 4% to other categories
- Flag which metrics are missing in the output so users understand data completeness

5. **Growth Metrics - Too Simplistic**

Problem: Using single-year growth rates is volatile and can be misleading.

Recommendation:

- Use **3-year or 5-year CAGR (Compound Annual Growth Rate)** instead of YoY
- Add **consistency score** - penalize volatile/erratic growth
- Consider **forward estimates** in addition to historical growth
- Add **market share growth** if data available

6. **Valuation - Context is Everything**

Critical Issues:

- P/E ratio scoring doesn't account for **interest rate environment** (low rates justify higher P/Es)
- No consideration of **growth-adjusted valuation** (PEG is included but needs more weight)
- Missing **Price-to-Sales (P/S)** and **Price-to-Free-Cash-Flow (P/FCF)** ratios

Recommendations:

- Add P/S ratio (especially important for unprofitable growth companies)
- Add P/FCF ratio (more reliable than P/E)
- Weight PEG ratio more heavily as it's superior to standalone P/E
- Consider **relative valuation** vs. sector median

7. **Profitability - Quality Matters**

Issue: Current system doesn't distinguish between sustainable vs. one-time profitability.

Add:

- **Operating Cash Flow to Net Income ratio** - measures earnings quality
- **Accruals ratio** - high accruals can signal accounting manipulation
- **Gross margin trend** - improving or deteriorating?

8. **Financial Health - Incomplete Picture**

Missing:

- **Cash-to-Debt ratio** - immediate debt coverage capability
- **Altman Z-Score** - bankruptcy prediction model
- **Times Interest Earned (TIE)** - you have "Interest Coverage" but ensure it's calculated correctly (EBIT/Interest Expense)

9. **Score Interpretation - Too Optimistic**

Problem: Your scale suggests 7.0-8.4 is "Strong Buy" but that might be too generous.

Recommendation:

- Recalibrate scale to be more conservative:
  - 9.0-10: Excellent (Strong Buy)
  - 7.5-8.9: Strong (Buy)
  - 6.0-7.4: Good (Hold/Cautious Buy)
  - 4.5-5.9: Below Average (Hold)

- Below 4.5: Weak/Poor (Reduce/Avoid)

10. **Temporal Issues**

Critical: The system uses static metrics, but fundamentals should show **trends**.

Add:

- **Trend indicators** for each metric (improving, stable, deteriorating)
- **Momentum scores** - give bonus points for metrics improving over time
- **YoY and 3-year comparison** for all major metrics

Specific Formula Issues

Current Ratio Scoring Problem:

You score Current Ratio > 2 as optimal, but **very high current ratios (>3)** might indicate poor asset management (excess cash not being deployed). Consider a bell curve approach.

P/E Scoring for Negative Earnings:

Giving score of 3 for negative P/E is reasonable, but doesn't differentiate between:

- A growth company investing heavily (acceptable)
- A failing company losing money (critical)

Solution: Check if the company is losing money with declining revenue (score 1-2) vs. investing for growth with rising revenue (score 4-5).

Major Recommendations Summary

Must-Have Changes:

1. **Add sector-specific benchmarks** (this is the #1 priority)
2. **Include Free Cash Flow metrics** prominently
3. **Fix missing data handling** - redistribute weights instead of neutral scores
4. **Use 3-5 year averages** for growth metrics, not single year
5. **Add trend/momentum indicators** to all categories
6. **Include earnings quality metrics**

Should-Have Changes:

1. Implement dynamic weighting based on company type
2. Add P/S and P/FCF to valuation
3. Add Altman Z-Score to financial health
4. Recalibrate interpretation scale to be more conservative

Nice-to-Have Changes:

1. Add forward-looking estimates where available

2. Include qualitative flags (e.g., "earnings beat estimates")
3. Peer comparison percentiles within sector

Proposed Enhanced Category Structure

Valuation (30%) → Keep weight but improve metrics:

- P/E (with sector context)
- P/B
- PEG (increase importance)
- EV/EBITDA
- **P/S** (add)
- **P/FCF** (add)

Profitability (30%) → Keep weight, add quality metrics:

- ROE
- ROA
- ROIC (add)
- Profit Margin
- Operating Margin
- **FCF Margin** (add)
- **Earnings Quality Score** (add - OCF/NI ratio)

Growth (20%) → Improve methodology:

- Revenue Growth (3-yr CAGR)
- Earnings Growth (3-yr CAGR)
- **FCF Growth** (add)
- **Growth Consistency** (add - penalize volatility)

Financial Health (15%) → Add depth:

- Current Ratio
- Quick Ratio
- Debt-to-Equity
- Interest Coverage
- **Altman Z-Score** (add)
- **Cash-to-Debt** (add)

Dividend (5%) → Keep but make conditional:

- Only apply full weight if company pays dividends
- Otherwise redistribute weight

Historical Context for Metrics - EXCELLENT Idea!

The Problem with Static Metrics:

A company with P/E = 25 could be:

- **Expensive** if its historical average P/E is 15
- **Cheap** if its historical average P/E is 40
- **Fairly valued** if 25 is its normal valuation

Recommended Implementation:

Add a **"Valuation Context Score"** or **"Historical Relative Score"** for key metrics:

Metrics That Benefit Most from Historical Context:

Critical for Historical Context:

1. **P/E Ratio** - most important
2. **P/B Ratio**
3. **EV/EBITDA**
4. **Dividend Yield** (high yield might indicate distress OR opportunity)
5. **Profit Margins** (are they improving or deteriorating?)
6. **Debt/Equity** (increasing leverage is concerning)

Less Critical but Still Useful:
7. ROE, ROA (trends matter)
8. Revenue/Earnings Growth (consistency matters)

Proposed Scoring Method:

For each key metric, calculate a **"Historical Percentile"**:

`Historical Percentile = Where current value ranks in last 3-5 years`

```
Example: P/E Ratio
- Current P/E: 18
- 5-year range: Low=12, High=35
- Current percentile: 25th percentile (in lowest quartile)
- Interpretation: Currently trading at attractive valuation relative to history
```

Historical Context Adjustment:

Add a **bonus/penalty modifier** to the base score:

For Valuation Metrics (lower is better):

- If current metric is in **bottom 25% of 5-year range**: +2 bonus points (historically cheap)
- If current metric is in **25th-50th percentile**: +1 bonus point (below average)
- If current metric is in **50th-75th percentile**: 0 adjustment (neutral)
- If current metric is in **top 25%**: -1 penalty point (historically expensive)

For Performance Metrics (higher is better - ROE, margins, etc.):

- Reverse the logic (top 25% gets bonus, bottom 25% gets penalty)

Example Calculation:

Company A:

- Base P/E Score: 6 (P/E = 28, moderately expensive by absolute standards)
- Historical Context: P/E is at 15th percentile of 5-year range (historically very cheap)
- Adjusted Score: 6 + 2 = **8** (now looks attractive)

Company B:

- Base P/E Score: 8 (P/E = 18, fairly valued by absolute standards)
- Historical Context: P/E is at 95th percentile of 5-year range (near all-time high)
- Adjusted Score: 8 - 1 = **7** (less attractive than it appears)

Specific Implementation Recommendations:

Priority Historical Metrics (Must-Have):

1. **P/E Ratio Historical Percentile**
   - Use 5-year lookback
   - Calculate monthly or quarterly P/E from historical data
   - Determine current percentile ranking
2. **Profit Margin Trend**
   - Are margins expanding (good) or contracting (bad)?
   - Compare current vs. 3-year average
   - Award bonus for improving trends
3. **Dividend Yield Context**
   - High yield could mean distressed company OR opportunity
   - Check if yield is high due to price drop (bad) or dividend increases (good)
   - Compare to 5-year average yield
4. **Debt/Equity Trend**
   - Is leverage increasing (concerning) or decreasing (good)?
   - Penalize rapidly increasing debt

Implementation Strategy:

Option 1: Simple Trend Indicators (Easier)

- Calculate 3-year average for each key metric
- Compare current value to average
- Award simple bonus/penalty: +1 if favorable, -1 if unfavorable, 0 if neutral

Option 2: Full Historical Percentile (Better but More Complex)

- Store 5 years of quarterly data for key metrics

- Calculate percentile ranking
- Use graduated bonus/penalty scale (-2 to +2 based on quartile)

Option 3: Hybrid (Recommended)

- Use full historical percentile for **most critical metrics** (P/E, P/B, margins)
- Use simple trend indicators for **secondary metrics** (debt trends, growth consistency)
- This balances accuracy with computational efficiency

Data Strategy with Yahoo Finance:

Since historical fundamentals aren't directly available as a time series, here's the approach:

For Historical Ratios:

javascript

```javascript
// Pseudo-code approach
1. Fetch quarterly/annual financial data for last 5 years
  - Use quoteSummary with 'earnings' module
  - Get historical price data

2. Calculate historical metrics:
  - Historical P/E = Historical Price / Historical EPS (quarterly)
  - Historical P/B = Historical Price / Historical Book Value per Share
  - Historical Margins from income statements

3. Store in your database:
  - Cache these historical calculations
  - Update quarterly (not daily, as fundamentals don't change that often)

4. Calculate percentile ranking:
  - Determine where current value falls in historical distribution
```

Database Schema Addition:

You'll want to store:

- **Company Historical Metrics Table:**
  - ticker, metric_name, value, date, percentile_rank
  - Update quarterly with new data
  - This avoids recalculating every time

Updated Scoring Framework with Historical Context:

Enhanced Valuation Category (30%):

Individual Metric Scoring:

1. Calculate **base score** using absolute thresholds (as in your current spec)
2. Calculate **historical percentile** (where current value ranks in 5-year range)
3. Apply **context adjustment** (+2 to -2 based on percentile for valuation metrics)
4. **Final Metric Score** = Base Score + Context Adjustment (capped at 0-10)

Example:

```
P/E Ratio Analysis:
- Current P/E: 22
- Base Score: 7 (falls in 20-25 range, moderately valued)
- 5-year P/E range: 15 to 45
- Current Percentile: 20th percentile (near historical low)
- Context Adjustment: +2
- Final P/E Score: 7 + 2 = 9 (excellent value in historical context)
```

Additional Benefit: Momentum Indicators

Historical data also enables **momentum/trend scores**:

- **Improving Fundamentals Bonus:** If ROE, margins, or revenue growth are all improving over past 3 years, add +0.5 to overall score
- **Deteriorating Fundamentals Penalty:** If debt is rising while profitability is falling, subtract -0.5
- **Mean Reversion Opportunity:** If valuation metrics are at historical extremes (bottom 10%), flag as potential opportunity

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My Recommendation:

Yes, absolutely implement historical context, with this phased approach:

Phase 1 (MVP):

- Add historical percentile scoring for **P/E, P/B, and Profit Margin** only
- Use simple 3-year comparison (current vs. 3-year average)
- Implement basic +1/0/-1 adjustment system

Phase 2 (Enhanced):

- Expand to full percentile ranking for all valuation metrics
- Add trend indicators for debt and profitability
- Implement -2 to +2 graduated scale

Phase 3 (Advanced):

- Add sector-relative historical context (not just company-specific)
- Implement momentum indicators
- Add mean-reversion flags for extreme valuations

Practical Considerations:

Computational Cost:

- Calculating historical percentiles adds processing time
- **Mitigation:** Cache historical calculations in database, update quarterly

Data Storage:

- Need to store 5 years of quarterly fundamentals per stock
- **Estimate:** ~20 metrics × 20 quarters × 1000 stocks = manageable database size

Update Frequency:

- Historical context doesn't need daily updates
- **Recommendation:** Update historical calculations when new quarterly earnings are released

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Would you like me to:

1. Create an updated functional specification incorporating historical context?
2. Provide sample code for calculating historical percentiles with yahoo-finance2?
3. Design a specific database schema for storing historical fundamental data?

Let me know which aspects you'd like to prioritize, and whether you want to start with Phase 1 (simple) or go directly to Phase 2 (full implementation).